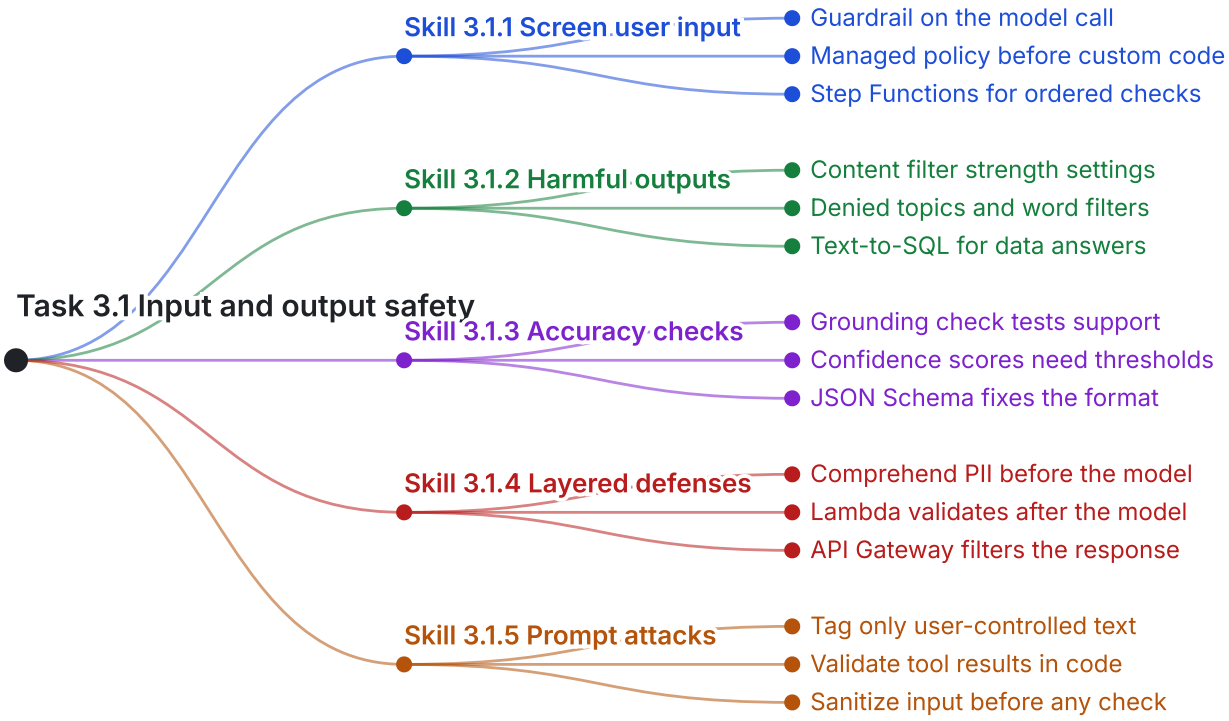
Memory sheet: Task 3.1 Implement input and output safety controls

Domain 3, weight 20 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Sensitive data already sits in S3 buckets	Amazon Macie for discovery and findings	A live chat message, since Macie does not sit in the request path
A live prompt or answer must be blocked or masked by policy	Bedrock Guardrails on the model call	Stored files, or tool results that the model-call guardrail skips
Exact PII spans or a redacted copy is required	Amazon Comprehend PII detection, with your code applying the change	A policy outcome such as refusing a topic, which is a guardrail job
A subject area must be refused, described in plain language	Denied topics with a natural language definition	A fixed list of exact names or identifiers, which needs word filters or a regex
Exact banned words or phrases must be blocked	Word filters with custom words or the managed profanity list	A rephrased insult, which content filters read for meaning
The answer must follow a supplied source passage	Contextual grounding check with the passage as the source	General knowledge questions with no source passage
Written eligibility rules must decide the answer, with an audit reason	Automated Reasoning checks with findings for your code	Prompt injection, off-topic text, or a streamed answer
Output must be a parseable object with fixed fields	JSON Schema structured output	A maximum string length, or proof that a value is true

Etch lines

1. Macie scans stored S3 objects, and never sits in a live request.
2. Guardrails returns a policy decision, and Comprehend returns entity locations.
3. Real-time Comprehend output lists locations, so your code must replace the spans.
4. A denied topic uses plain language, and a word filter uses exact terms.
5. A score blocks nothing until code applies a threshold.
6. Wrap only user-controlled text in input tags for the prompt attack filter.

Flash questions

- Q: Which control scans S3 buckets for sensitive data that already sits at rest? A: Amazon Macie, since it works on stored objects and not on live messages.
- Q: Which guardrail policy type refuses a subject described in plain language? A: Denied topics, since they use a natural language definition.
- Q: Does the content filter strength setting apply to a list of banned product names? A: No, strength applies to categories, so use word filters for exact names.
- Q: Which Comprehend mode fits a live chat turn that needs PII locations now? A: Real-time analysis of one document, with your code replacing the spans.
- Q: Does a guardrail on the model call check a tool result inside the request? A: No, toolResult content is not evaluated, so screen it with ApplyGuardrail or code.
- Q: Which three inputs does a contextual grounding check need? A: The source passage, the user query, and the model response to guard.

- Q: Does a schema-valid object prove that its values are true? A: No, a schema checks shape only, so use grounding or Automated Reasoning for facts.
- Q: What does an empty outputs list from ApplyGuardrail mean? A: No action was taken, so an empty list is not a block.

Source: Task 3.1 sections for Skill 3.1.1 to 3.1.5, Guardrails versus Comprehend versus Macie versus filters, Read a long question, Distractor patterns, and Drill. AWS docs and the study guide sections above, no practice-test text.